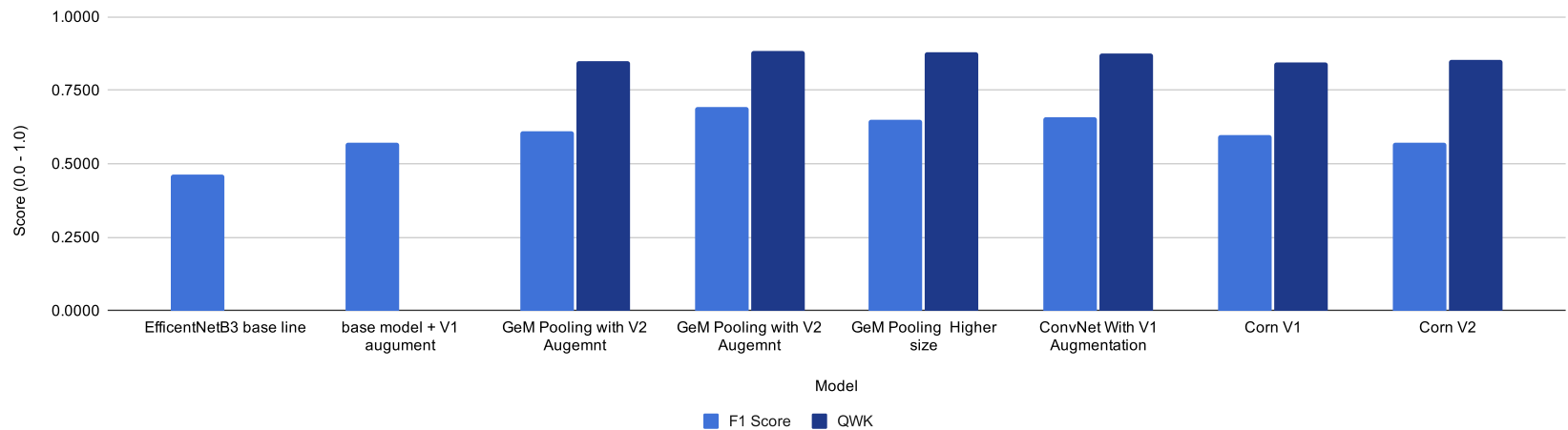


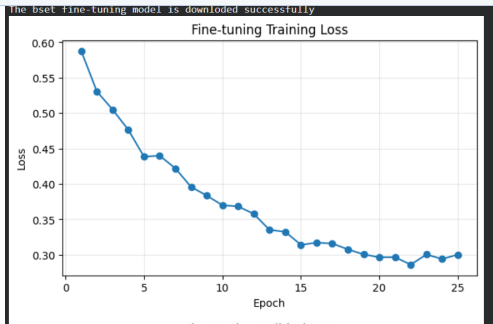
Model Benchmark & Comparison							
TOTAL MODELS		HIGHEST F1 SCORE		HIGHEST QWK SCORE		TOP PERFORMING MODEL	
8		0.6919		0.8841		GeM Pooling with V2 Augemnt	
Architectures Evaluated		Validation Dataset		Quadratic Weighted Kappa		Highest Overall QWK	
Model Name	Architecture	Description	F1 Score	QWK	Train Loss	Validation Loss	Epochs
EfficentNetB3 base line	With Freezing		0.4610			0.6500	10
base model + V1 augment	With Freezing	Used Focal Loss	0.5730		0.4400	0.3000	10
GeM Pooling with V2 Augemnt	With Freezing	Note that is was less PreProcessing	0.6125	0.8495		0.3620	40
GeM Pooling with V2 Augemnt	No Freeze	Image size=300, 70% for training	0.6919	0.8841	0.3000	0.2404	40
GeM Pooling Higher size	No Freeze Better General	Image size=384, 80% for training, 5e-4 WD	0.6488	0.8774	0.2500	0.2900	40
ConvNet With V1 Augmentation	With Freezing		0.6580	0.8733		0.2830	40
Corn V1	With Freezing	V1 augment	0.5988	0.8430		0.2989	20
Corn V2	With Freezing	V2 augemnt with Gem layer	0.5735	0.8532		0.2841	20



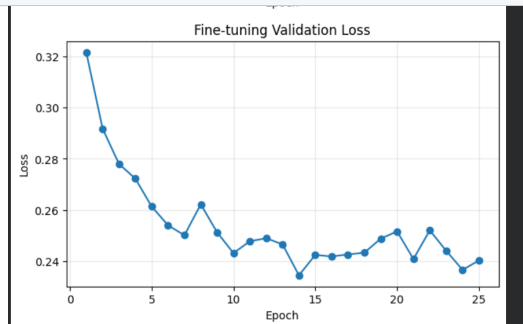
Model Loss Curves & Training Diagnostics								
<i>Training and validation loss curves for comparative analysis</i>								

MODEL 1: GeM Pooling with V2 Augemnt (Candidate)

Training Loss Curve

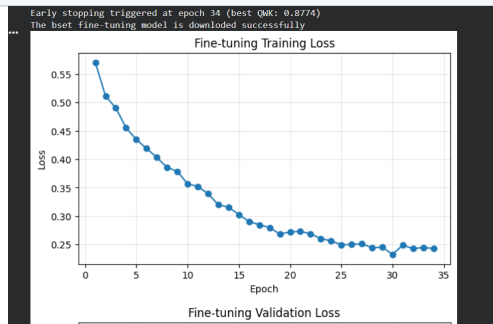


Validation Loss Curve



MODEL 2: GeM Pooling Higher size (Champion)

Training Loss Curve



Validation Loss Curve

